

A Term Project Report On

Smart Blind Stick

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Electronics and Communication

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CERTIFICATE

This is to certify that the practical/term work carried out on the subject of the project titled “SMART BLIND STICK” and recorded in this report is the bona fide work of Ms. Lakhani Dhruvi Girishbhai(EC007) of B.Tech Semester VI in the branch of Electronics and Communication Engineering during the academic year 2025–2026.

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ABSTRACT

The Smart Blind Stick is an assistive device designed to enhance the safety and mobility of visually impaired individuals. This system integrates multiple sensors and communication technologies to overcome the limitations of traditional white canes. An ultrasonic sensor is used to detect obstacles by measuring distance using the time-of-flight principle, while a water sensor detects wet or uneven surfaces to prevent slipping hazards. The ESP32 microcontroller processes all sensor data in real time and generates appropriate alerts through a buzzer and a vibration motor, where vibration intensity varies according to the proximity of obstacles.

For emergency situations, a push-button-based SOS feature is implemented. When activated, the system acquires the user's real-time location using a GPS module, which operates based on trilateration technique. This location data is transmitted using a LoRa module that employs Chirp Spread Spectrum modulation, enabling long-range and low-power communication without relying on internet connectivity. At the receiver side, the data is further sent to a guardian's smartphone via WiFi using secure HTTPS protocol through a Telegram bot, allowing instant notification and location tracking through Google Maps.

The system combines multiple communication protocols such as UART for GPS, SPI for LoRa, ADC for analog sensing, and PWM for vibration control, ensuring efficient and reliable operation. The Smart Blind Stick provides real-time obstacle detection, hazard alerts, and emergency communication, making it a cost-effective and practical solution for improving independence and safety of visually impaired individuals.

BACKGROUND

1. Motivation

Visually impaired individuals face significant challenges while navigating independently in their daily lives. Traditional white canes help detect obstacles on the ground, but they are limited in their ability to detect objects at different heights, sudden obstacles, or hazardous conditions such as water puddles or pits. This limitation increases the risk of accidents, especially in crowded, unfamiliar, or uneven environments.

Another major concern is the lack of an effective emergency communication system. In critical situations, visually impaired individuals may not be able to inform their guardians about their location or condition. This delay in communication can lead to serious consequences.

The motivation behind this project is to design a smart, reliable, and user-friendly assistive device that enhances safety and independence. By integrating sensors, real-time alerts, and long-range wireless communication, the Smart Blind Stick aims to provide a safer navigation experience and quick emergency response support.

2. Literature Review

Several assistive technologies have been developed in recent years to support visually impaired individuals. Early systems primarily focused on obstacle detection using ultrasonic sensors and provided alerts through sound or vibration. While these systems improved basic navigation, they lacked advanced features such as real-time tracking and long-range communication.

Later developments introduced GPS-based tracking systems combined with GSM modules to send location data to guardians. However, GSM-based systems depend heavily on network availability and may not function effectively in remote or rural areas. Similarly, systems using Bluetooth or Wi-Fi have limited communication range and are not suitable for long-distance applications.

Recent advancements in wireless communication technologies have introduced LoRa (Long Range) communication, which provides low-power and long-distance data transmission without relying on internet connectivity. This makes LoRa highly suitable for assistive devices that need reliable communication even in areas with poor network coverage.

Based on these studies, the proposed Smart Blind Stick integrates ultrasonic sensing, water detection, GPS tracking, and LoRa communication to overcome the limitations of existing systems and provide a more robust and reliable solution.

3. Basic Theory

The Smart Blind Stick operates on the principle of sensing, processing, decision-making, and communication.

The ultrasonic sensor detects obstacles by transmitting ultrasonic waves and measuring the time taken for the echo to return. This method is based on the concept of Time of Flight, which allows accurate distance calculation.

A water sensor is used to detect the presence of water or wet surfaces. It works on the principle of electrical conductivity, where the presence of water changes the resistance and voltage level, which is then converted into a digital value using the ESP32's ADC.

The ESP32 microcontroller acts as the central processing unit, receiving data from sensors and making decisions based on predefined thresholds. Depending on the situation, it activates a buzzer for audio alerts and a vibration motor for tactile feedback. The vibration intensity is controlled using Pulse Width Modulation (PWM), allowing different levels of feedback based on obstacle distance.

For location tracking, the GPS module uses the principle of Trilateration to determine the exact position of the user in terms of latitude and longitude.

In emergency situations, when the SOS button is pressed, the location data is transmitted using a LoRa module. LoRa uses Chirp Spread Spectrum, which enables long-distance and low-power communication. At the receiver side, the data is sent to the guardian's smartphone using WiFi and HTTPS protocol, allowing real-time location tracking through a map interface.

Overall, the system integrates multiple sensing and communication technologies to provide an efficient, reliable, and real-time assistive solution for visually impaired individuals.

PROBLEM DEFINITION & DESIGN

PROBLEM STATEMENT

Traditional mobility aids such as white canes provide only basic obstacle detection and are limited to ground-level awareness. They are unable to detect obstacles at different distances, such as poles, vehicles, or overhanging objects, which increases the risk of accidents for visually impaired individuals. Additionally, these conventional devices cannot detect hazardous conditions like water puddles or pits, which may lead to slipping or falling.

Another major limitation is the absence of an emergency communication mechanism. In critical situations, visually impaired individuals are unable to share their location with guardians or family members, resulting in delayed assistance. Existing solutions that use GSM or internet-based communication often depend on network availability, making them unreliable in remote or low-connectivity areas.

Therefore, there is a need for an intelligent assistive system that can:

- Detect obstacles at various distances
- Identify water or unsafe surfaces
- Provide real-time alerts through sound and vibration
- Enable long-range communication for emergency situations
- Share accurate location information with guardians

The proposed Smart Blind Stick addresses these challenges by integrating sensors, microcontroller-based processing, and LoRa communication technology to enhance user safety and independence.

SYSTEM DESIGN

The Smart Blind Stick is designed as an integrated system consisting of sensing, processing, alert, and communication units. The ESP32 microcontroller acts as the central control unit, coordinating all operations in real time.

1. Input Section (Sensors)

The input section includes sensors that continuously monitor the environment:

- **Ultrasonic Sensor:**

Detects obstacles in front of the user by measuring distance using the Time of Flight principle. It helps identify objects that are not detectable by a traditional stick.

- **Water Sensor:**

Detects the presence of water or wet surfaces based on changes in electrical conductivity. The analog signal is converted into a digital value using the ESP32's ADC.

- **GPS Module:**

Provides real-time location data using Trilateration. This information is used during emergency situations to track the user's location.

2. Processing Unit

- **ESP32 Microcontroller:**

Acts as the brain of the system. It receives sensor data, processes it using predefined logic, and decides the appropriate action. It supports multiple communication protocols such as UART (GPS), SPI (LoRa), ADC (water sensor), and GPIO (sensors and actuators), making it suitable for this integrated application.

3. Output Section (Alert System)

The system provides alerts to the user through:

- **Buzzer:**

Generates audio alerts when an obstacle is very close or when a dangerous situation is detected.

- **Vibration Motor:**

Provides tactile feedback. The intensity of vibration is controlled using Pulse Width Modulation (PWM), allowing the user to understand the distance of obstacles.

4. Communication Section

- **LoRa Module (Transmitter & Receiver):**

Used for long-range wireless communication. It transmits emergency alerts and location data using Chirp Spread Spectrum, which ensures reliable communication even in areas without internet connectivity.

- **WiFi Module (ESP32):**

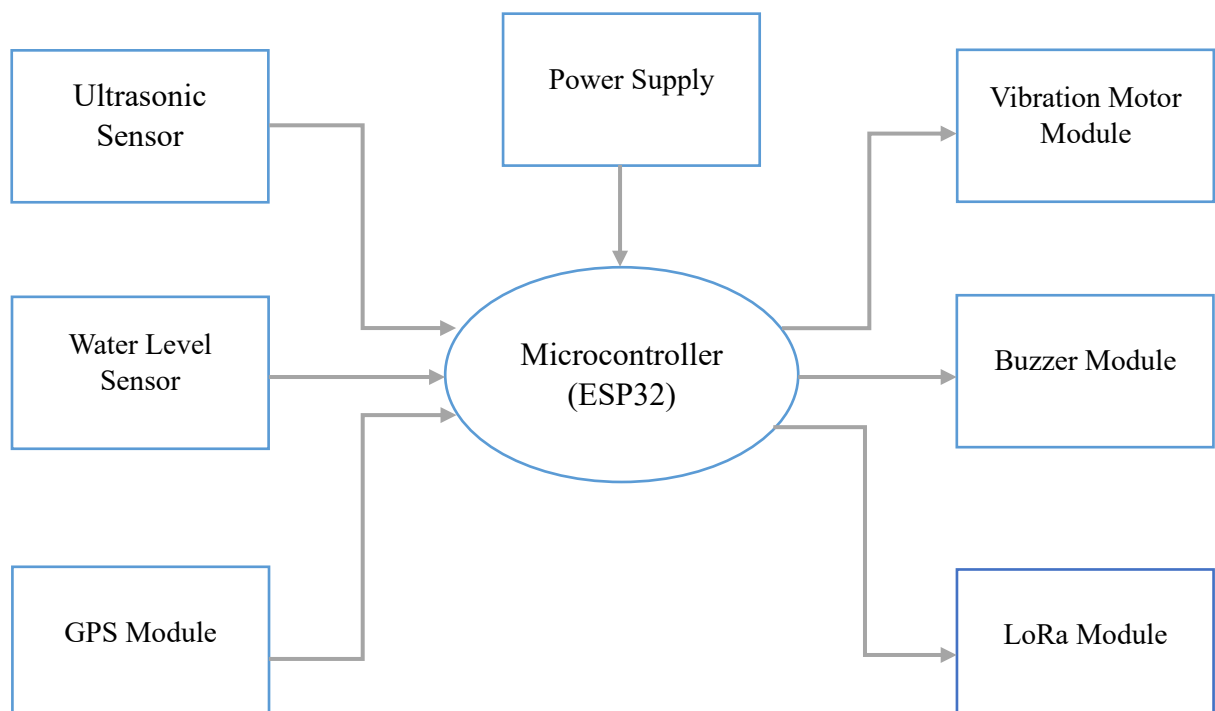
At the receiver side, WiFi is used to send data to the cloud or messaging platform (e.g., Telegram) using HTTPS protocol, enabling real-time notification to the guardian's smartphone.

5. Power Supply

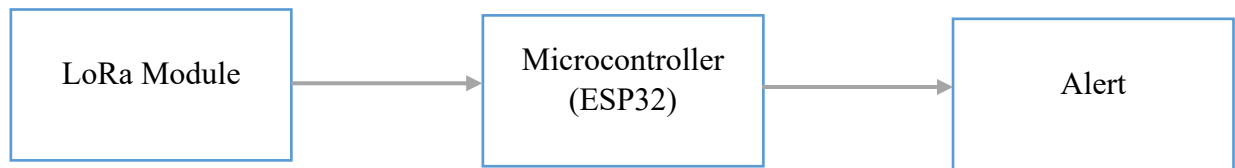
A rechargeable battery powers the entire system, ensuring portability and ease of use. The system is designed to consume low power, making it suitable for continuous operation.

BLOCK DIAGRAM

Transmitter Block

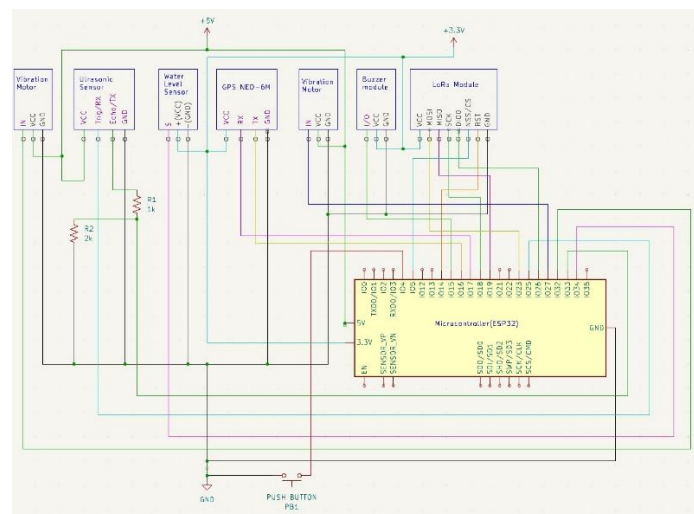


Receiver Block

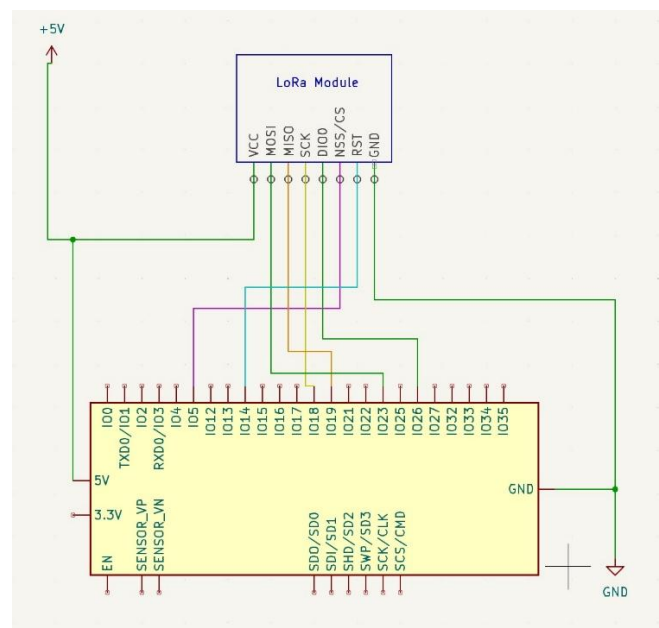


Circuit Diagram :

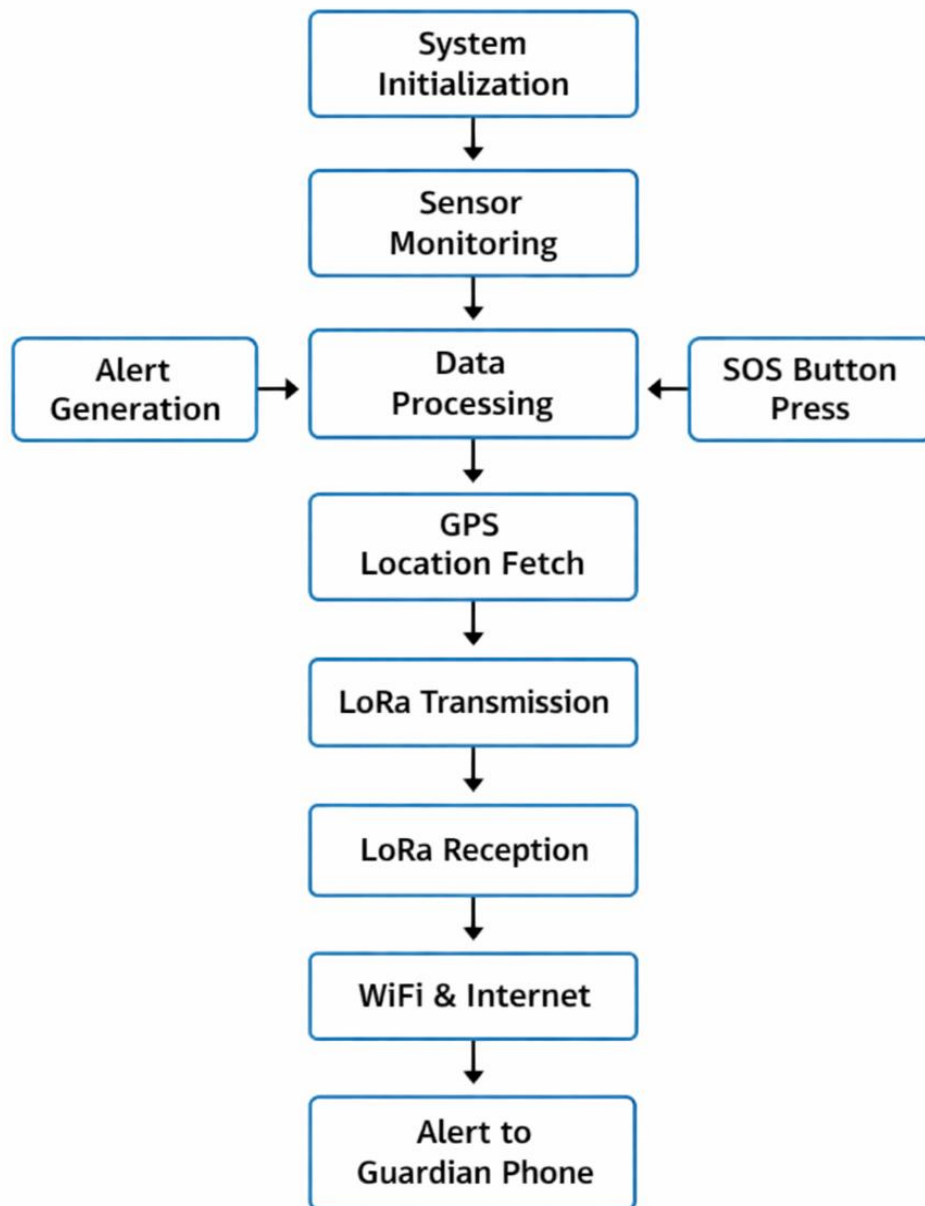
Tansmitter :



Receiver:



Workflow:



TEST SETUP AND METHODOLOGY

Hardware Components

1. ESP32 Microcontroller

The ESP32 acts as the central control unit of the system. It continuously receives data from the ultrasonic sensor and water sensor and processes this data to make real-time decisions. Based on the sensor readings, it activates the buzzer and vibration motor to alert the user. It also interfaces with the GPS module and LoRa module for location tracking and wireless communication.

- **Specifications:**

Operating Voltage: 3.3V

CPU: Dual-core processor

ADC Resolution: 12-bit (0–4095)

Built-in WiFi and Bluetooth

- **Technology Used:**

SAR ADC, Embedded WiFi

- **Protocol Used:**

UART (GPS), SPI (LoRa), ADC (Water Sensor), GPIO

- **Libraries Used:**

WiFi.h, SPI.h

- **Reason for Selection:**

Supports multiple communication protocols and real-time processing in a single compact device.

2. Ultrasonic Sensor (HC-SR04)

The ultrasonic sensor detects obstacles in front of the user at different distances. It works by sending ultrasonic waves and measuring the time taken for the echo to return.

- **Specifications:**

Operating Voltage: 5V

Range: 2–400 cm

Frequency: 40 kHz

- **Technology Used:**

Time of Flight

- **Protocol Used:**

Digital GPIO (Trigger and Echo pins)

- **Reason for Selection:**

Provides accurate distance measurement and helps detect obstacles not reachable by a normal stick.

3. Active Buzzer Module

The buzzer provides audio alerts when obstacles are very close or when a danger condition is detected.

- **Specifications:**

Operating Voltage: 3.3V–5V

Type: Active buzzer

- **Technology Used:**

Piezoelectric sound generation

- **Protocol Used:**

Digital GPIO

- **Reason for Selection:**

Simple and effective for generating alert sounds.

4. Vibration Motor Module

The vibration motor provides tactile feedback to the user. The vibration intensity varies based on obstacle distance.

- **Specifications:**

Operating Voltage: 3V–5V

- **Technology Used:**

Pulse Width Modulation (PWM)

- **Protocol Used:**

PWM (ledcWrite in ESP32)

- **Libraries Used:**

No external library (ESP32 built-in PWM)

- **Reason for Selection:**

Useful for visually impaired users in noisy environments.

5. LoRa Module (Ra-02)

The LoRa module is used for long-range wireless communication between transmitter and receiver.

- **Specifications:**

Frequency: 433 MHz

Range: Up to 10–15 km

Low power consumption

- **Technology Used:**

Chirp Spread Spectrum

- **Protocol Used:**

SPI

- **Libraries Used:**

LoRa.h

- **Reason for Selection:**

Provides long-distance communication without internet dependency.

6. GPS Module

The GPS module provides real-time location information of the user.

- **Specifications:**

Update Rate: 1–5 Hz

Cold Start: ~38 seconds

- **Technology Used:**

Trilateration

- **Protocol Used:**

UART

- **Libraries Used:**

TinyGPS++

- **Reason for Selection:**

Enables accurate location tracking during emergencies.

7. Water Sensor

The water sensor detects wet surfaces or pits to prevent slipping.

- **Specifications:**
Operating Voltage: 3–5V
- **Technology Used:**
Analog sensing
- **Protocol Used:**
ADC
- **Libraries Used:**
No library required
- **Reason for Selection:**
Provides safety against water hazards.

8. LoRa Antenna

The antenna is used to transmit and receive RF signals effectively.

- **Specifications:**
Frequency: 433 MHz
- **Reason for Selection:**
Improves communication range and signal strength.

9. Power Supply Unit

A rechargeable battery powers the entire system.

- **Specifications:**
Type: Li-ion battery
Output: 3.7/5V
- **Reason for Selection:**
Ensures portability and continuous operation.

Methodology:

1. Initialization

- ESP32 is initialized.
- Pins for ultrasonic sensor, water sensor, buzzer, and vibration motor are configured.
- GPS and LoRa modules are started for communication.

2. Obstacle Detection (Ultrasonic Sensor)

- The ultrasonic sensor sends sound waves and receives the echo.
- Distance is calculated using:

$$D = \frac{T \times 0.034}{2}$$

Where, 0.034 cm/ μ s is speed of sound

3. Distance-Based Decision

Based on calculated distance:

- Distance > 80 cm
→ No alert (Safe)
- 50 cm < Distance $\leq$ 80 cm
→ Low vibration
- 25 cm < Distance $\leq$ 50 cm
→ Medium vibration
- 10 cm < Distance $\leq$ 25 cm
→ Strong vibration
- Distance $\leq$ 10 cm
→ Strong vibration + Buzzer ON (Critical alert)

4. Water Detection

The water sensor gives an analog voltage, which is converted into a digital value using the ESP32's ADC.

ADC Concept

- ESP32 ADC is typically 12-bit resolution
- Digital output range: 0 to 4095
- Corresponding voltage range: 0V to 3.3V

- Conversion Formula:

$$DigitalValue = \frac{v_{in}}{v_{ref}} * 4095$$

Where: v_{in} = input voltage from sensor, $v_{ref} = 3.3V$

- Value < 500 → Dry
- 500 – 2000 → Low water
- 2000 – 3000 → Medium water
- > 3000 → High water (Danger)
- If water detected → Alert using vibration/buzzer

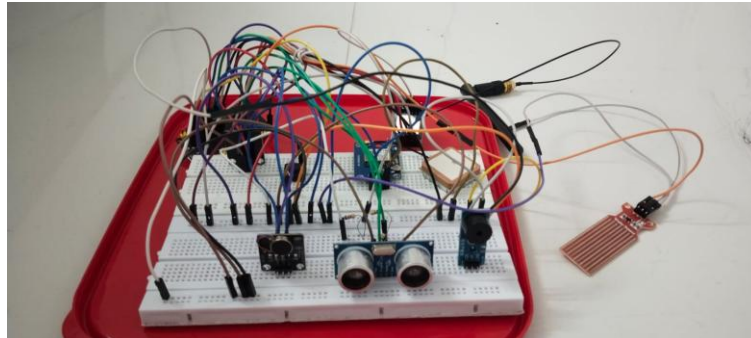
5. GPS Location Tracking

- GPS module provides: Latitude, Longitude
- This is sent during emergency if SOS button Pressed

6. Alert Generation

- Buzzer → Audio alert for critical danger
- Vibration Motor → Intensity changes with distance

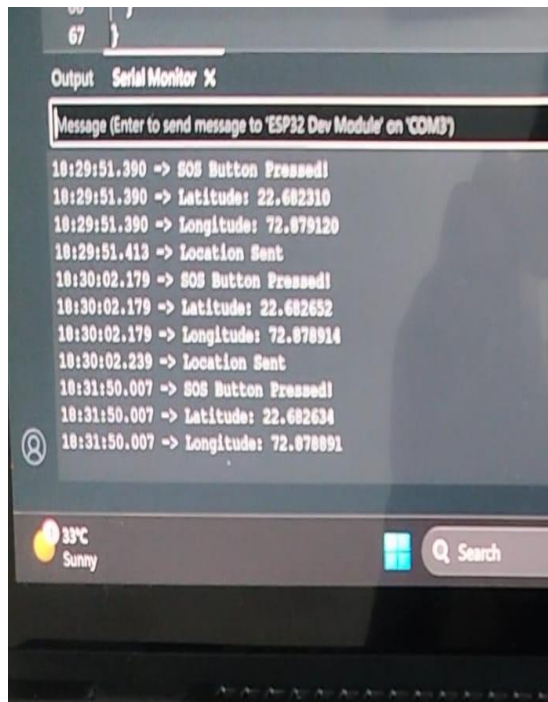
Setup:



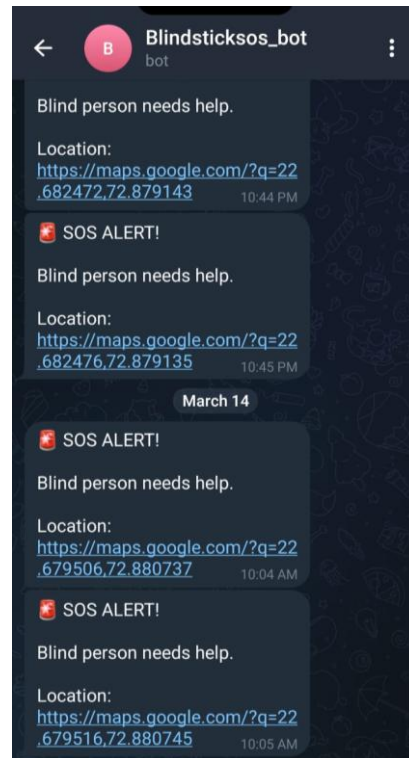
RESULT AND DISCUSSION

```
The water sensor value: 0
The water sensor value: 0
The water sensor value: 0
The water sensor value: 25
The water sensor value: 196
The water sensor value: 587
The water sensor value: 625
The water sensor value: 1434
The water sensor value: 1449
The water sensor value: 1454
The water sensor value: 1469
```

Readings of Water level Sensor



Location sending when sos button is pressed



location received on telegram

CONCLUSION

The Smart Blind Stick is an effective solution designed to improve the safety and independence of visually impaired individuals. It successfully overcomes the limitations of traditional canes by detecting obstacles and water hazards in real time. The system provides immediate alerts through sound and vibration, helping users navigate safely. The integration of GPS enables location tracking, while the LoRa module ensures long-range communication during emergencies. This allows guardians to receive alerts even in areas with poor network connectivity. The use of low-power components makes the device efficient and portable for daily use. Overall, the project demonstrates how modern technology can be used to assist people with disabilities. It enhances confidence, mobility, and quick response in critical situations. This system can be further improved and expanded for real-world applications.

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